

UKMARS Gemini V2 PiPico Pin assignments

Chassis and Mezzanine Pins

Pico GPIO	Mezz Net ID	Used on	Device	Device name	Description
	0 GPIO00	Mezz	J2-2	Breakout connector	UART0-TX for bluetooth or GPIO pin
	1 GPIO01	Mezz	J2-3	Breakout connector	UART0-RX for bluetooth or GPIO pin
	2 LMTR1	Chassis	U1-7	DRV8833 motor controller	Left motor control signal 1
	3 LMTR2	Chassis	U1-8	DRV8833 motor controller	Left motor control signal 2
	4 RMTR1	Chassis	U1-11	DRV8833 motor controller	Right motor control signal 1
	5 RMTR2	Chassis	U1-12	DRV8833 motor controller	Right motor control signal 2
	6 RQA	Chassis	J2-3	Right motor quadrature encoder	Signal A
	7 RQB	Chassis	J2-2	Right motor quadrature encoder	Signal B
	8 LQA	Chassis	J1-3	Left motor quadrature encoder	Signal A
	9 LQB	Chassis	J2-2	Left motor quadrature encoder	Signal B
	10 GPIO10	Mezz	J5-4	Breakout connector	I2C1-SDA for display or GPIO pin
	11 GPIO11	Mezz	J5-3	Breakout connector	I2C1-SCL for display or GPIO pin
	12 GPIO12	Mezz	D1	Indicator LED	Left side
	13 GPIO13	Mezz	D2	Indicator LED	Right side
	14 GPIO14	Mezz	SW2	Pushbutton	Right side
	15 GPIO15	Mezz	SW1	Pushbutton	Left side

Line Sensor Pins

Pico GPIO	Mezz Net ID	Used on	Device	Device name	Description
	16 SDA	Line Sensor	Not used		
	17 SCL	Line Sensor	Not used		
	18 SEN4DIO	Line Sensor	Q6	BC337-40	Radius emitter driver
	19 SEN3DIO	Line Sensor	D6	Indicator LED	S/F right side indicator
	20 SEN2DIO	Line Sensor	D5	Indicator LED	Line central indicator
	21 SEN1DIO	Line Sensor	D4	Indicator LED	Radius left side indicator
	22 SEN0DIO	Line Sensor	Q1	BC337-40	Line and start/finish emitter driver
	26 SEN2AIP	Line Sensor	Q4	BPW85C	Line right side detector
	27 SEN1AIP	Line Sensor	Q2 and Q5	BPW85C	Radius and Start/finish detectors
	28 SEN0AIP	Line Sensor	Q3	BPW85C	Line left side detector
Not available	SEN3AIP	Not available			
Not available	SEN4AIP	Not available			
Not available	SEN5AIP	Not available			

Wall Sensor Pins

Pico GPIO	Mezz Net ID	Used on	Device	Device name	Description
	16 SDA	Wall Sensor	Not used		
	17 SCL	Wall Sensor	Not used		
	18 SEN4DIO	Wall Sensor	D3	Indicator LED	Right side indicator
	19 SEN3DIO	Wall Sensor	D2	Indicator LED	Central (to the right) indicator
	20 SEN2DIO	Wall Sensor	D1	Indicator LED	Left side indicator
	21 SEN1DIO	Wall Sensor	Q2	BC337-40	Driver for diagonal emitters
	22 SEN0DIO	Wall Sensor	Q1	BC337-40	Driver for front emitter
	26 SEN2AIP	Wall Sensor	RDPD1	BPW85C	Right diagonal detector
	27 SEN1AIP	Wall Sensor	FPD1	BPW85C	Front detector
	28 SEN0AIP	Wall Sensor	LDPD1	BPW85C	Right diagonal detector
Not available	SEN3AIP	Not available			
Not available	SEN4AIP	Not available			
Not available	SEN5AIP	Not available			